Steam EDA

Patterns & Trends

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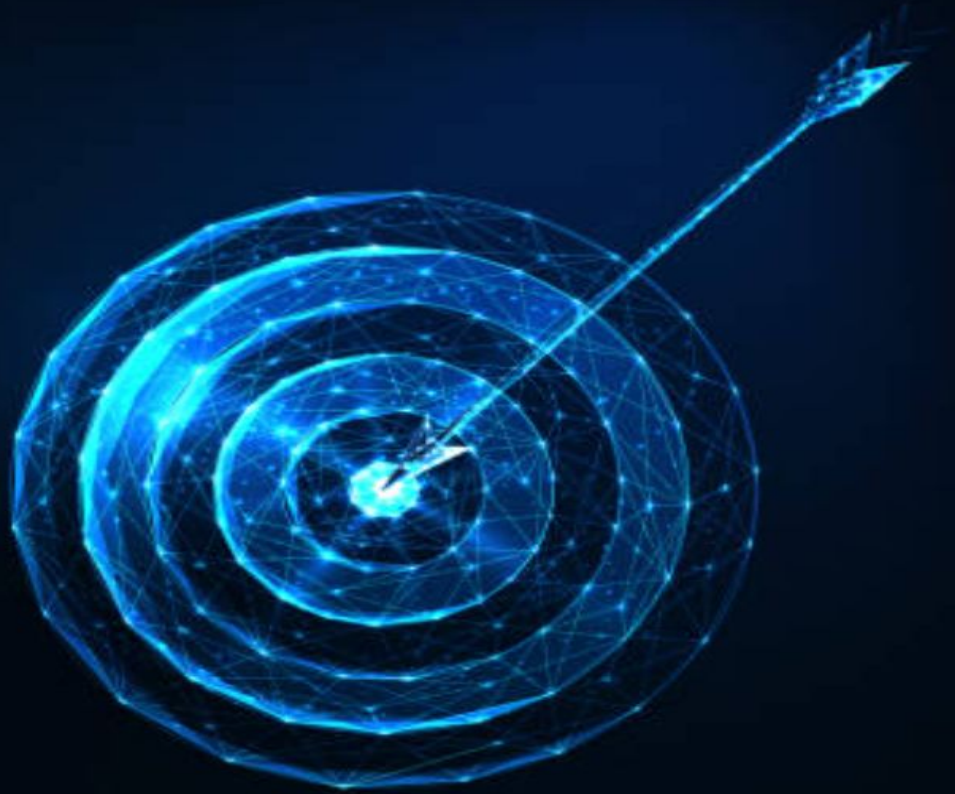
The **ULTIMATE** goal for this project is to understand why some games become global phenomenon, by looking at trends and patterns within the 2014 Steam dataset found on kaggle.



*Due to the size of the dataset, analysis consists of a sample of 5,000 entries.

Price VS Owners

Exploring data to uncover what makes a Steam game popular

QUESTION

Does the price of a game have a significant impact on how many people will own it.

KEY FINDINGS

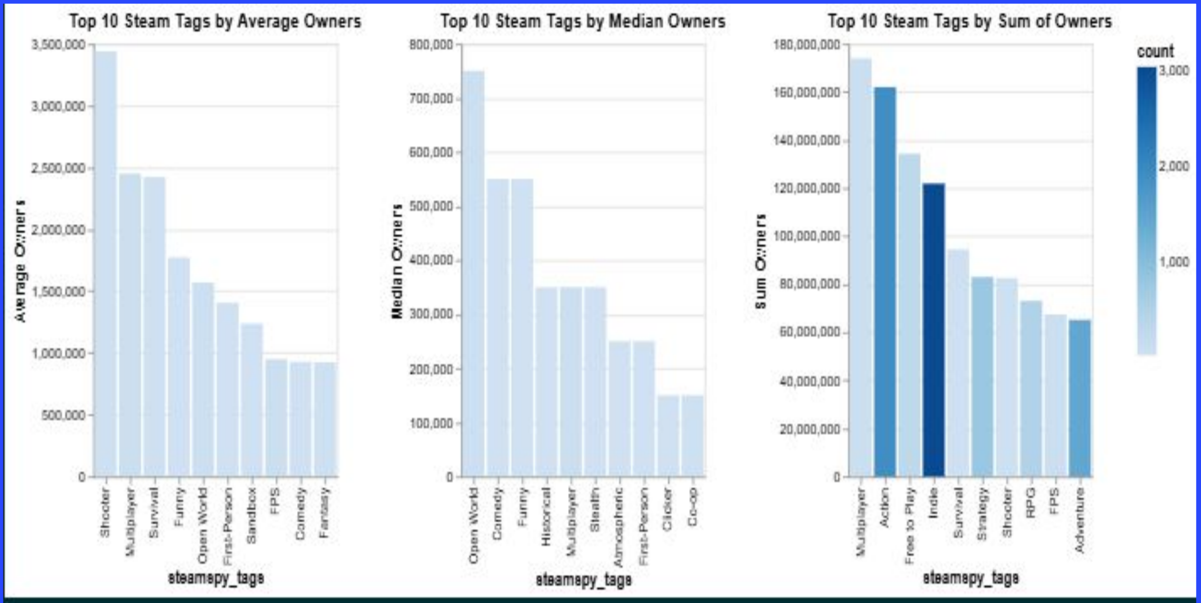
- Most Games are free and low amount of players
- Appears to be a weak negative correlation between owners and price
- Few games with over 1 million owners very few over 4 million
- Very few games over 50$

TAKEAWAY

Higher prices seems to slightly reduce game purchases

Tags VS Owners

Exploring data to uncover what makes a Steam game popular



QUESTION

Which Tags (type of game) are commercially most successful

KEY FINDINGS

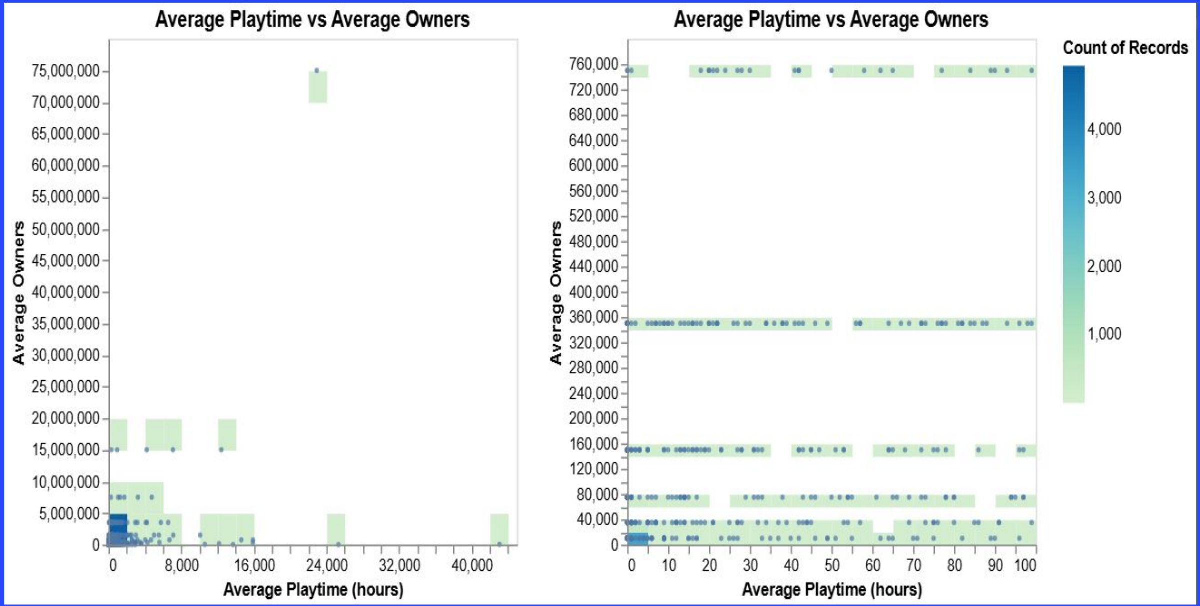
- Each Graph only have about 2-3 common tags as another, indicating a fairly skewed dataset. Some Games may skyrocket a Tags average while its median stays very low compared to other tags
- **Multiplayer** was the only tag in all three graphs, while **openworld**, **first person**, and **comedy/funny** showed up in both top mean and median

TAKEAWAY

It seems Multiplayer is the most consistently successful tag

Avg Playtime VS Owners

Exploring data to uncover what makes a Steam game popular



QUESTION

Do games people play longer become more popular

KEY FINDINGS

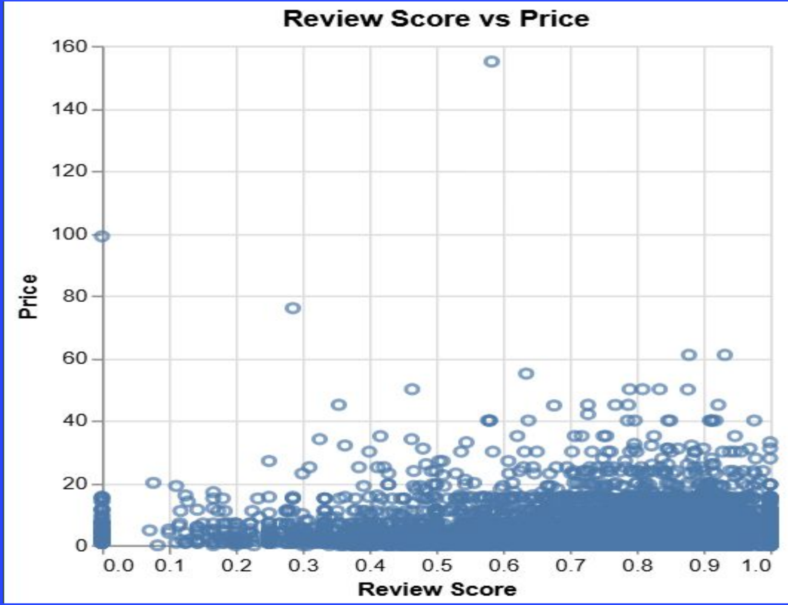
- Most Games are very low plates and very low playtime
- appears to be a very small negative correlation, but appears to be other variables
- Avg Owners may be binned estimates as they are all in straight lines
- outliers on both axis and entire rows of less extreme outliers

TAKEAWAY

Very high mean playtime may slightly decrease popularity

Review Score VS Price

Exploring data to uncover what makes a Steam game popular



*There are two columns for reviews indicating number of positive and negative reviews for a game. Review score is calculated: positive/(positive+negative)

QUESTION

Does paying more buy a better game

KEY FINDINGS

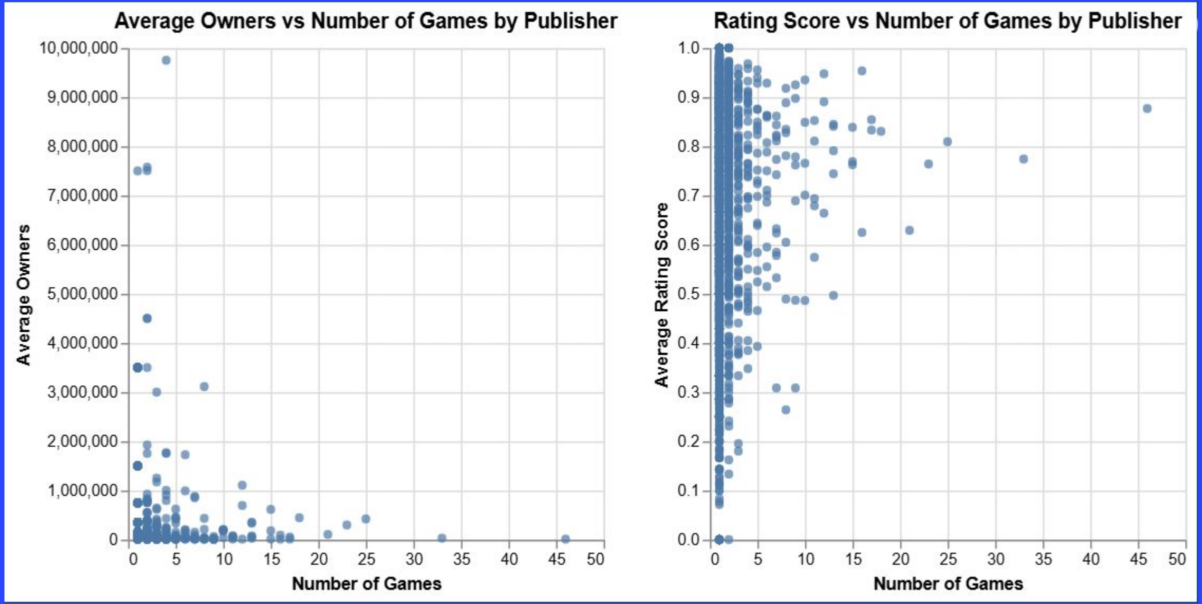
- Seems to be a slight positive correlation
- 65$ seems to be the highest price while maintaining a high review score
- Seems to be a cluster of games with a score between 0.7 and 0.95 with a price between 20-30$
- Cluster of 0 score between 0-20$, possibly from unrated games

TAKEAWAY

A higher price (under 65$) may indicate a better game

Published Games VS Owners and Ratings

Exploring data to uncover what makes a Steam game popular



QUESTION

Will releasing more games make a publisher more successful

KEY FINDINGS

- With more games published we see a decrease in average owners, but we see an increase in the average rating score
- Publishers with 1 Game have the highest variability in both owners and ratings making them high risk high reward games

TAKEAWAY

More games, higher ratings, fewer owners per game

CONCLUSIONS

Key takeaways from our exploratory data analysis

OVERALL SUMMARY

Price is not a primary driver of popularity.

Multiplayer tags correlate with success.

Frequent publishers release higher-rated, but not necessarily more popular, titles.

Most successful games remain affordable ($50–65).

High average playtime does not guarantee a larger player base.

NEXT STEPS

- **Explore additional variables such as release date, developer, supported platforms, or genres.**
- **Build a predictive model to estimate game popularity from multiple features.**
- **Compare patterns across different Steam datasets or more recent years.**

FINAL TAKEAWAY

Commercial success appears to depend on a combination of game quality, genre, and player appeal rather than any single variable.